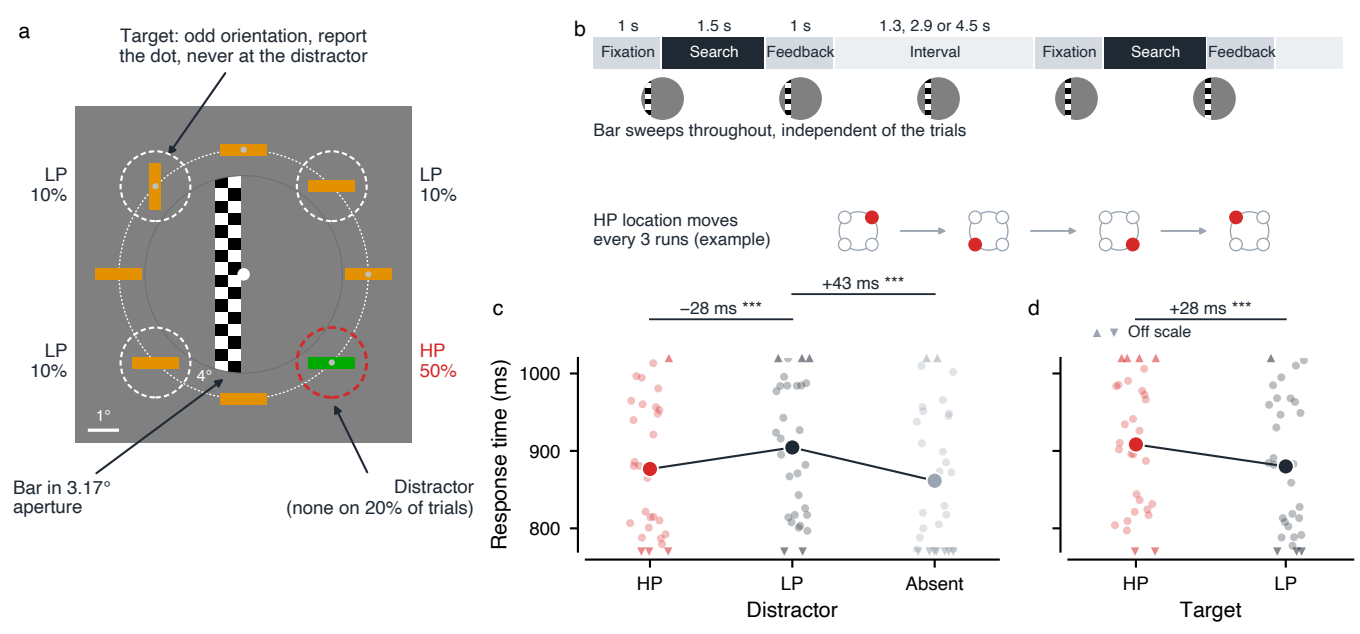
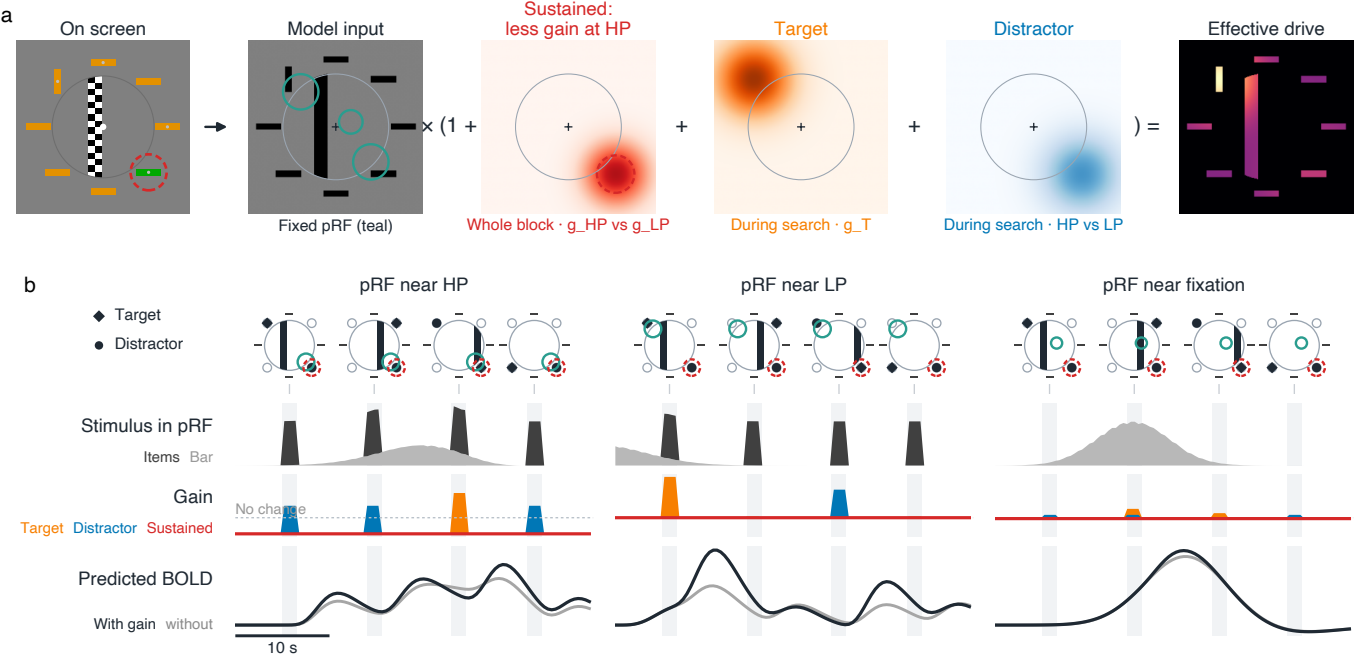
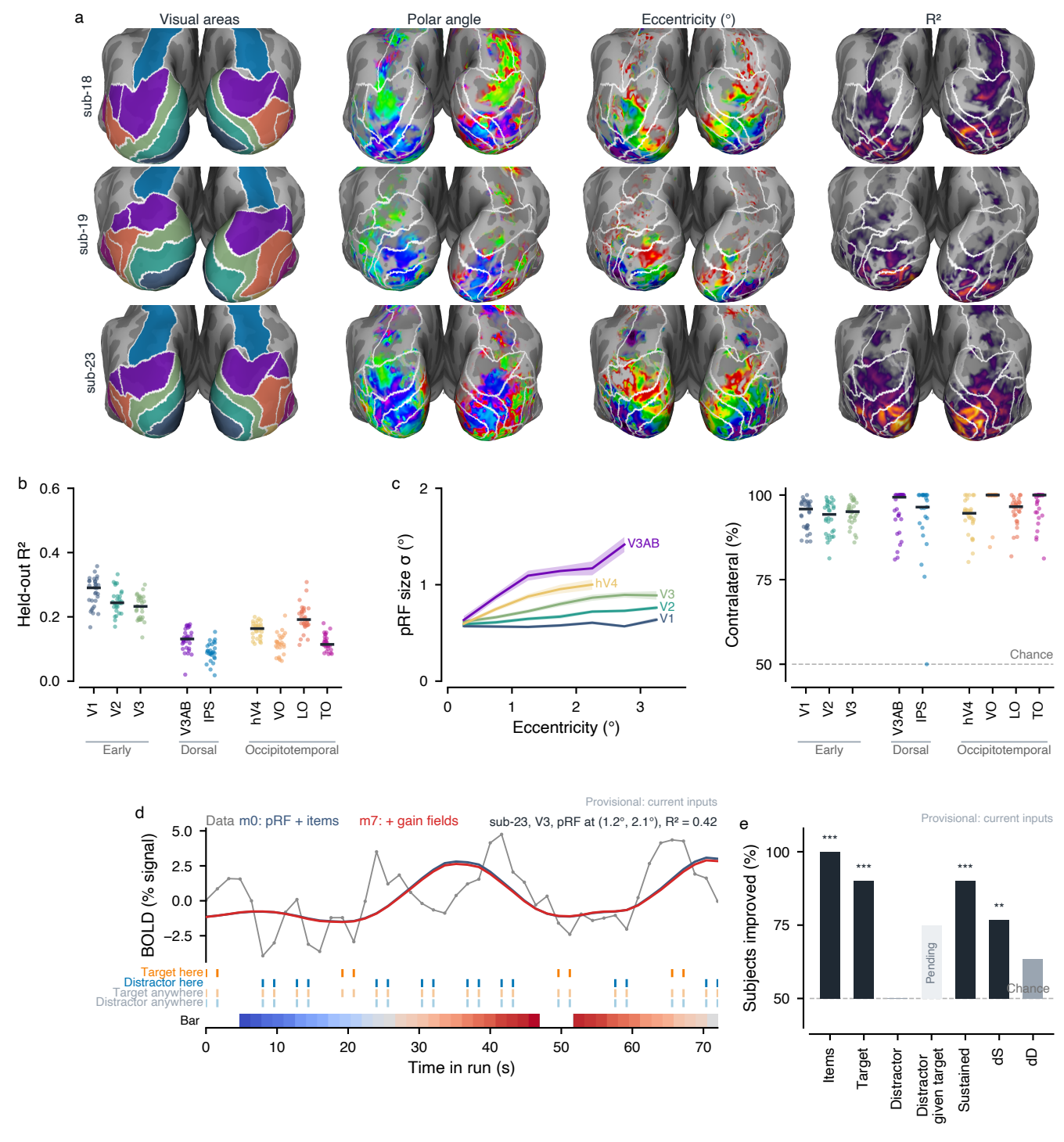
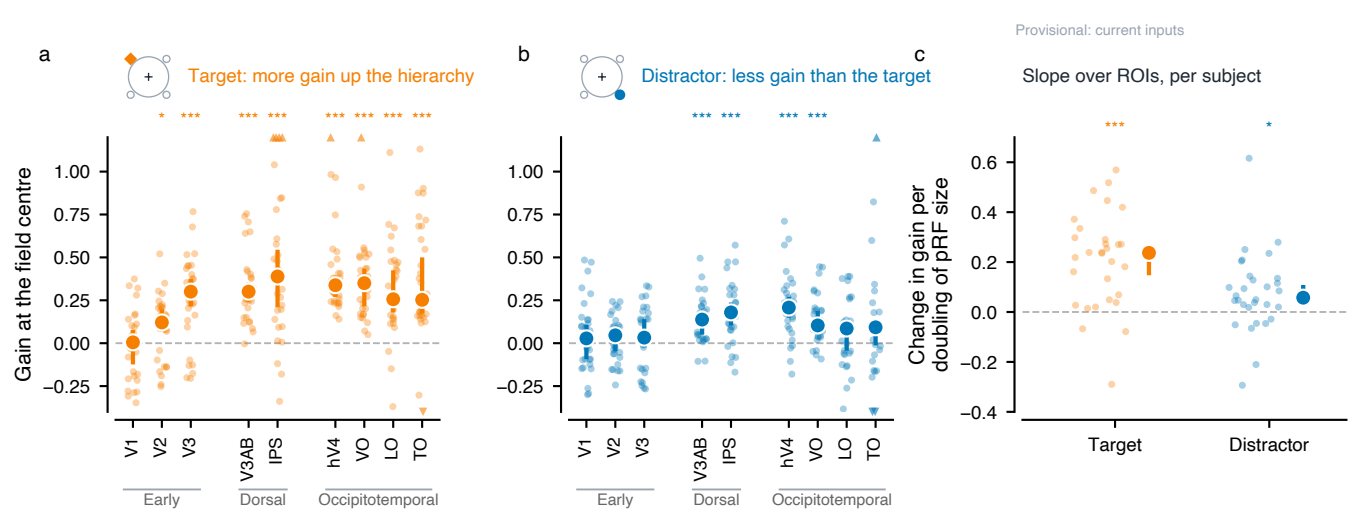
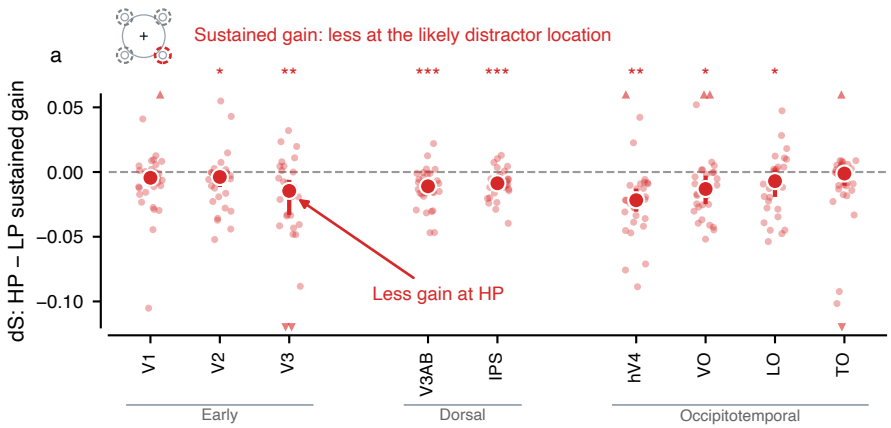
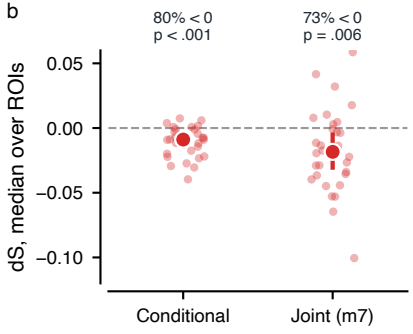

a
Target: odd orientation, report the dot, never at the distractor
LP 10%
LP 10%
LP 10%
HP 50%
4°
1°
Bar in 3.17° aperture
Distractor (none on 20% of trials)
b
1 s
1.5 s
1 s
1.3, 2.9 or 4.5 s
Fixation
Search
Feedback
Interval
Fixation
Search
Feedback
Bar sweeps throughout, independent of the trials
HP location moves every 3 runs (example)
c
–28 ms ***
+43 ms ***
Response time (ms)
1000
900
800
HP
LP
Absent
Distractor
d
+28 ms ***
Off scale
HP
LP
Target

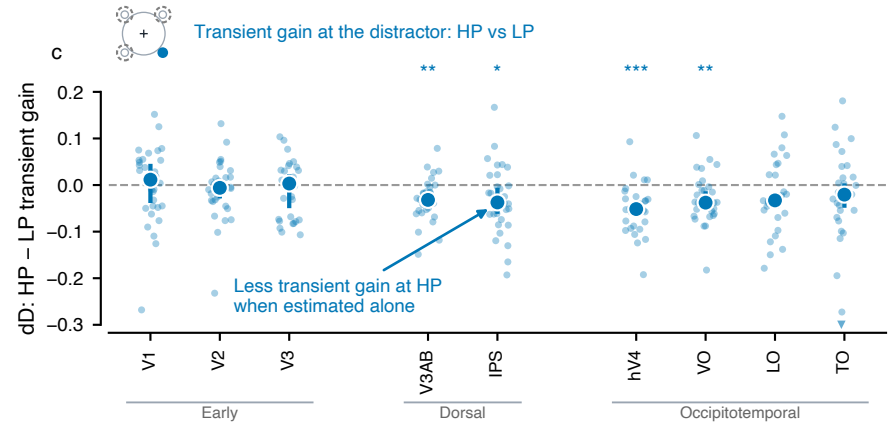

a
On screen
Model input
Fixed pRF (teal)
× (1 +
Sustained:
less gain at HP
Whole block · g_HP vs g_LP
+
Target
During search · g_T
+
Distractor
During search · HP vs LP
) =
Effective drive
b
Target
Distractor
pRF near HP
pRF near LP
pRF near fixation
Stimulus in pRF
Items Bar
Gain
Target Distractor Sustained
No change
Predicted BOLD
With gain without
10 s

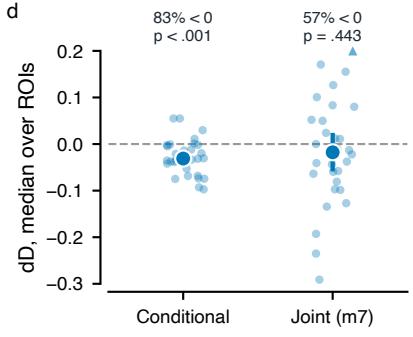

a
Visual areas
Polar angle
Eccentricity (°)
R²
sub-18
sub-19
sub-23
b
Held-out R²
0.6
0.4
0.2
0.0
V1
V2
V3
V3AB
IPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal
c
pRF size σ (°)
2
1
0
V3AB
hV4
V3
V2
V1
0
1
2
3
Eccentricity (°)
Contralateral (%)
100
75
50
Chance
V1
V2
V3
V3AB
IPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal
d
Provisional: current inputs
Data m0: pRF + items
m7: + gain fields
sub-23, V3, pRF at (1.2°, 2.1°), R² = 0.42
BOLD (% signal)
5.0
2.5
0.0
−2.5
Target here
Distractor here
Target anywhere
Distractor anywhere
Bar
0
10
20
30
40
50
60
70
Time in run (s)
e
Provisional: current inputs
Subjects improved (%)
100
75
50
***
***
***
**
Pending
Chance
Items
Target
Distractor
Distractor given target
Sustained
dS
dD

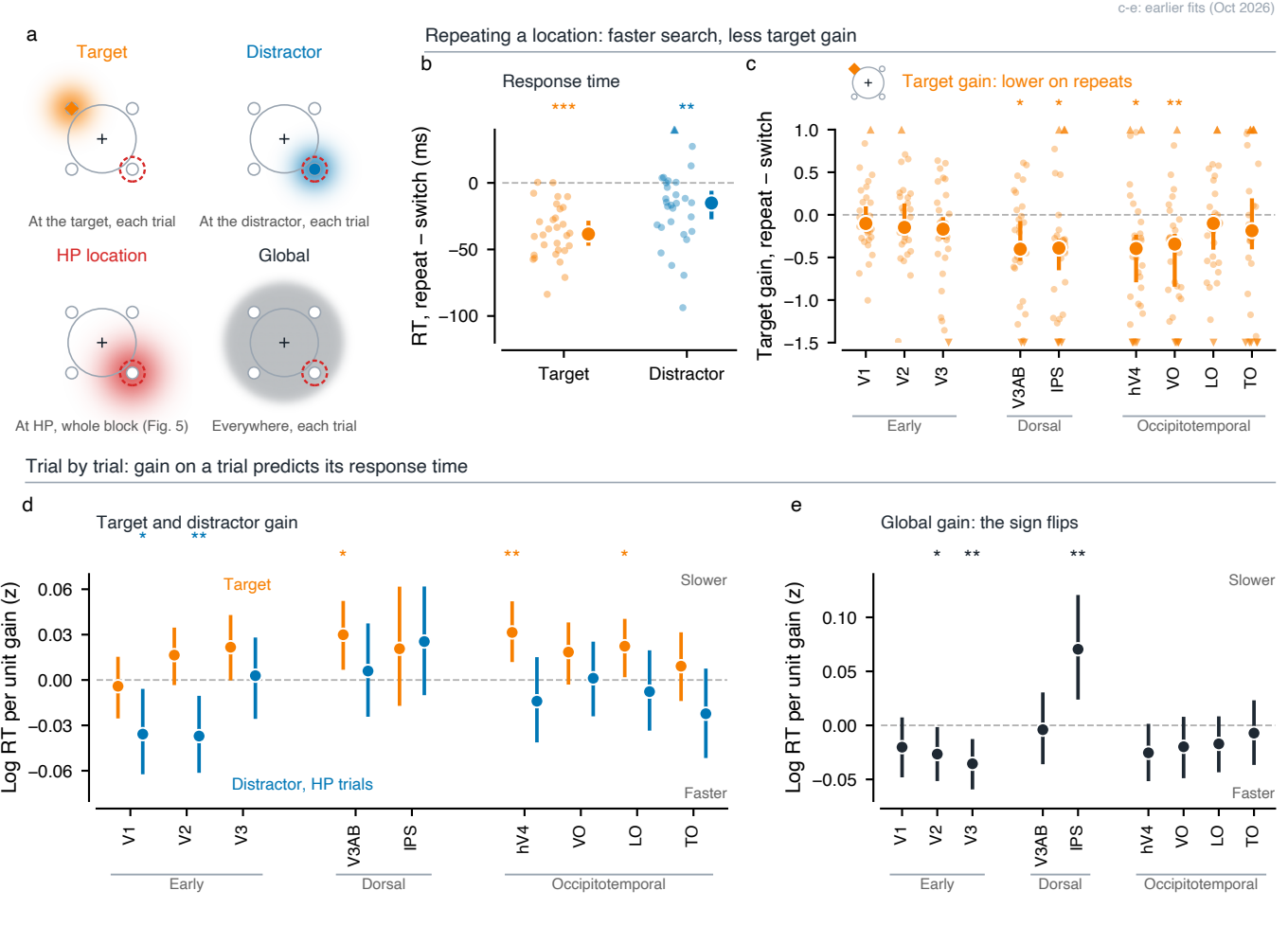

Provisional: current inputs
a
Target: more gain up the hierarchy
Gain at the field centre
1.00
0.75
0.50
0.25
0.00
−0.25
V1
V2
V3
V3AB
IPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal
b
Distractor: less gain than the target
c
Slope over ROIs, per subject
Change in gain per doubling of pRF size
0.6
0.4
0.2
0.0
−0.2
−0.4
Target
Distractor

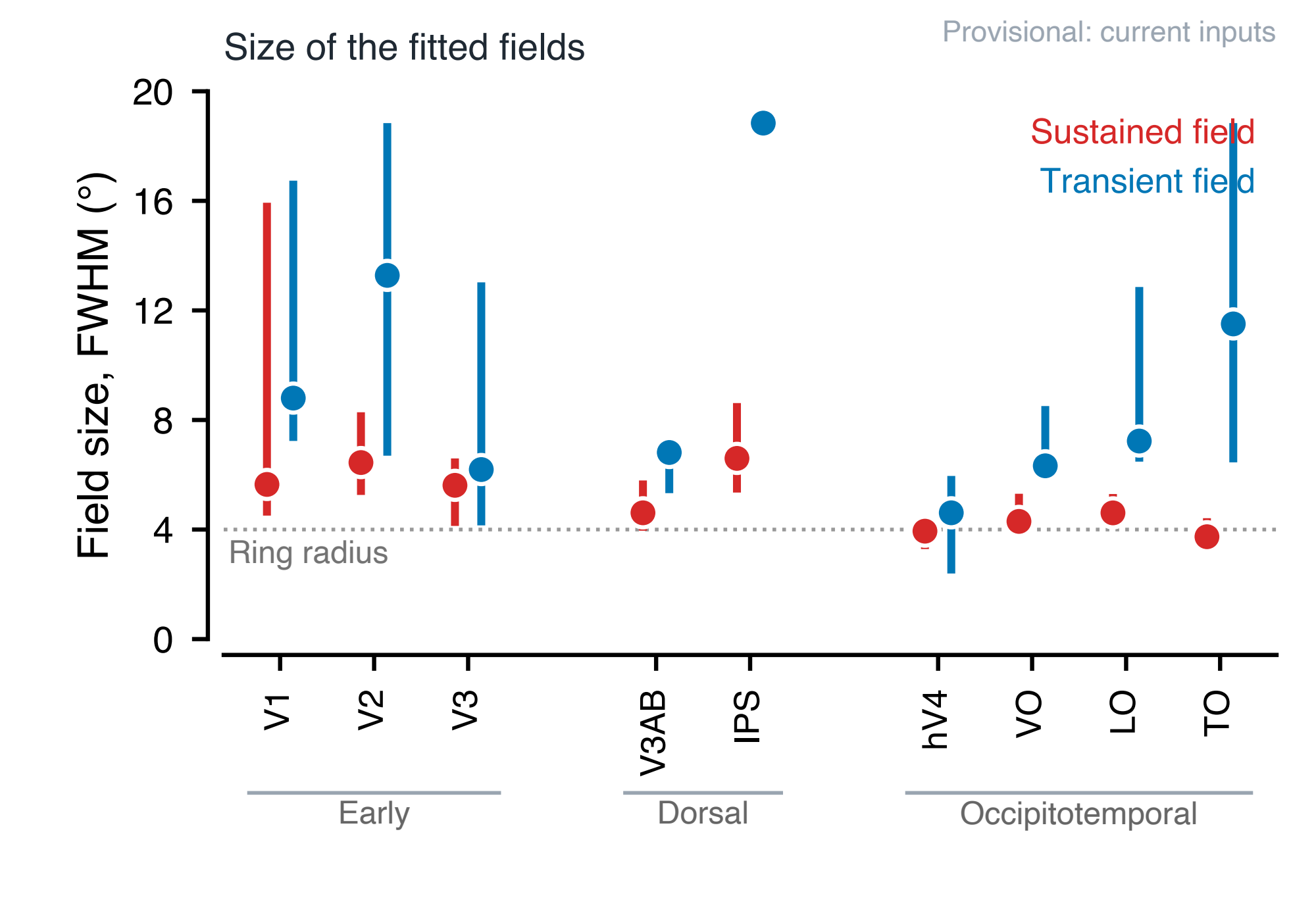

a
Sustained gain: less at the likely distractor location
dS: HP – LP sustained gain
0.05
0.00
−0.05
−0.10
* ** *** *** ** * *
Less gain at HP
V1 V2 V3 V3AB IPS hV4 VO LO TO
Early
Dorsal
Occipitotemporal

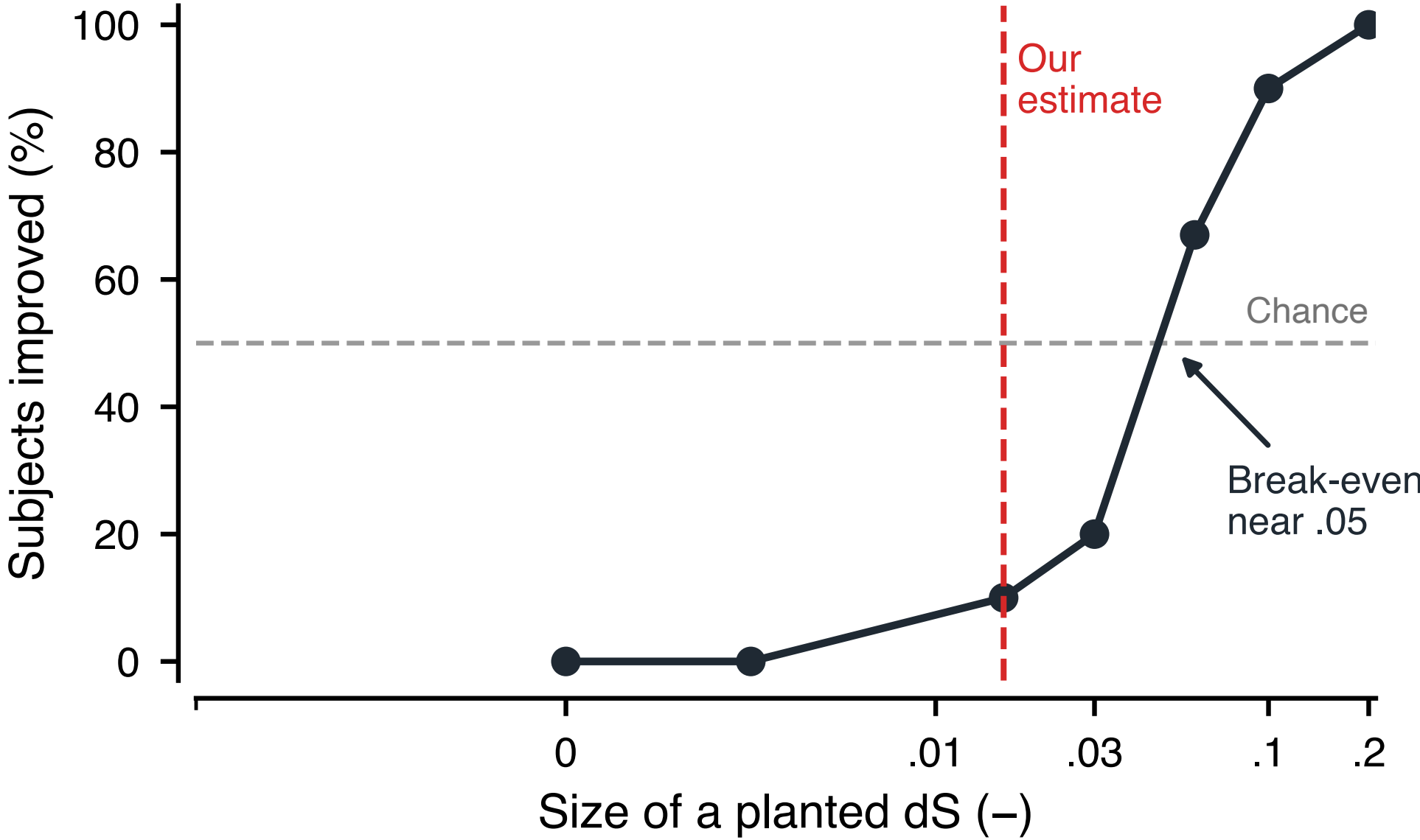

b
80% < 0
p < .001
73% < 0
p = .006
dS, median over ROIs
0.05
0.00
−0.05
−0.10
Conditional
Joint (m7)

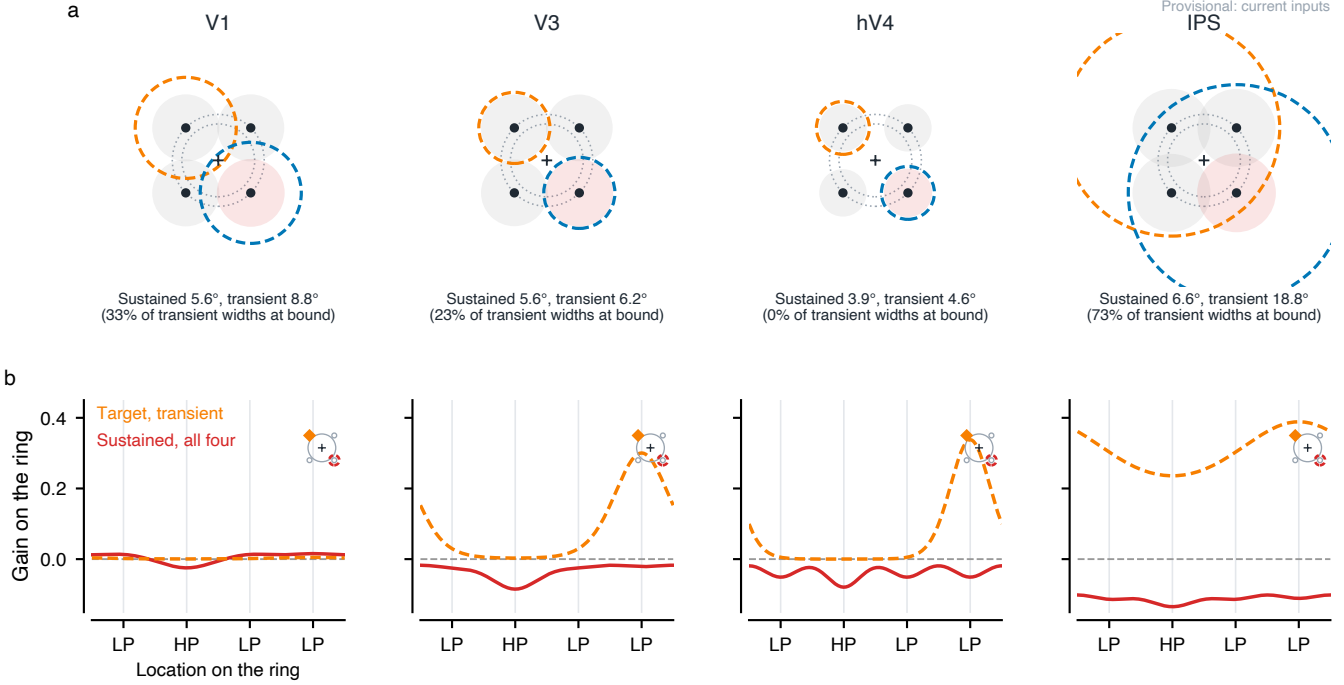

c
Transient gain at the distractor: HP vs LP
dD: HP – LP transient gain
0.2
0.1
0.0
−0.1
−0.2
−0.3
** * *** **
Less transient gain at HP
when estimated alone
V1 V2 V3 V3AB IPS hV4 VO LO TO
Early
Dorsal
Occipitotemporal


d
83% < 0
p < .001
57% < 0
p = .443
dD, median over ROIs
0.2
0.1
0.0
−0.1
−0.2
−0.3
Conditional
Joint (m7)


a
Target
Distractor
At the target, each trial
At the distractor, each trial
HP location
Global
At HP, whole block (Fig. 5)
Everywhere, each trial
Repeating a location: faster search, less target gain
b
Response time
RT, repeat – switch (ms)
Target
Distractor
c
Target gain: lower on repeats
Target gain, repeat – switch
V1
V2
V3
V3AB
IPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal
Trial by trial: gain on a trial predicts its response time
d
Target and distractor gain
Log RT per unit gain (z)
Target
Distractor, HP trials
Slower
Faster
e
Global gain: the sign flips


Size of the fitted fields
Provisional: current inputs
Field size, FWHM (°)
20
16
12
8
4
0
Sustained field
Transient field
Ring radius
V1
V2
V3
V3AB
IPS
hV4
VO
LO
TO
Early
Dorsal
Occipitotemporal


100
80
60
40
20
0
Subjects improved (%)
Our
estimate
Chance
Break-even
near .05
0
.01
.03
.1
.2
Size of a planted dS (–)


a
V1
V3
hV4
IPS
Provisional: current inputs
Sustained 5.6°, transient 8.8°
(33% of transient widths at bound)
Sustained 5.6°, transient 6.2°
(23% of transient widths at bound)
Sustained 3.9°, transient 4.6°
(0% of transient widths at bound)
Sustained 6.6°, transient 18.8°
(73% of transient widths at bound)
b
Target, transient
Sustained, all four
Gain on the ring
0.4
0.2
0.0
LP
HP
LP
LP
Location on the ring